

SIADH in the Elderly: A Complication of Pneumonia

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Introduction

Community-acquired pneumonia (CAP) is the most frequent infectious cause of hospitalization and mortality in industrialized countries. The incidence of hospitalization due to CAP is about 3 cases per 1,000 adults and those that are hospitalized have a mortality rate around 10%. Hyponatremia is the most common electrolyte imbalance seen in clinical practice. Studies have shown that on average, 4% of patients presenting with pneumonia will have hyponatremia due to SIADH. Hyponatremia in patient with pneumonia has been proven to be a poor prognostic indicator – associated with increased hospital mortality, prolonged hospital stays and increased rates of readmission. Geriatric patients are at the highest risk for developing hyponatremia. Emerging evidence suggests that the presence of hyponatremia in pneumonia may serve as a marker for systemic inflammation and cytokine dysregulation. Correcting hyponatremia in pneumonia patients has been shown to reduce the risk of severe complications, but aggressive correction can also pose risks, particularly in the elderly.

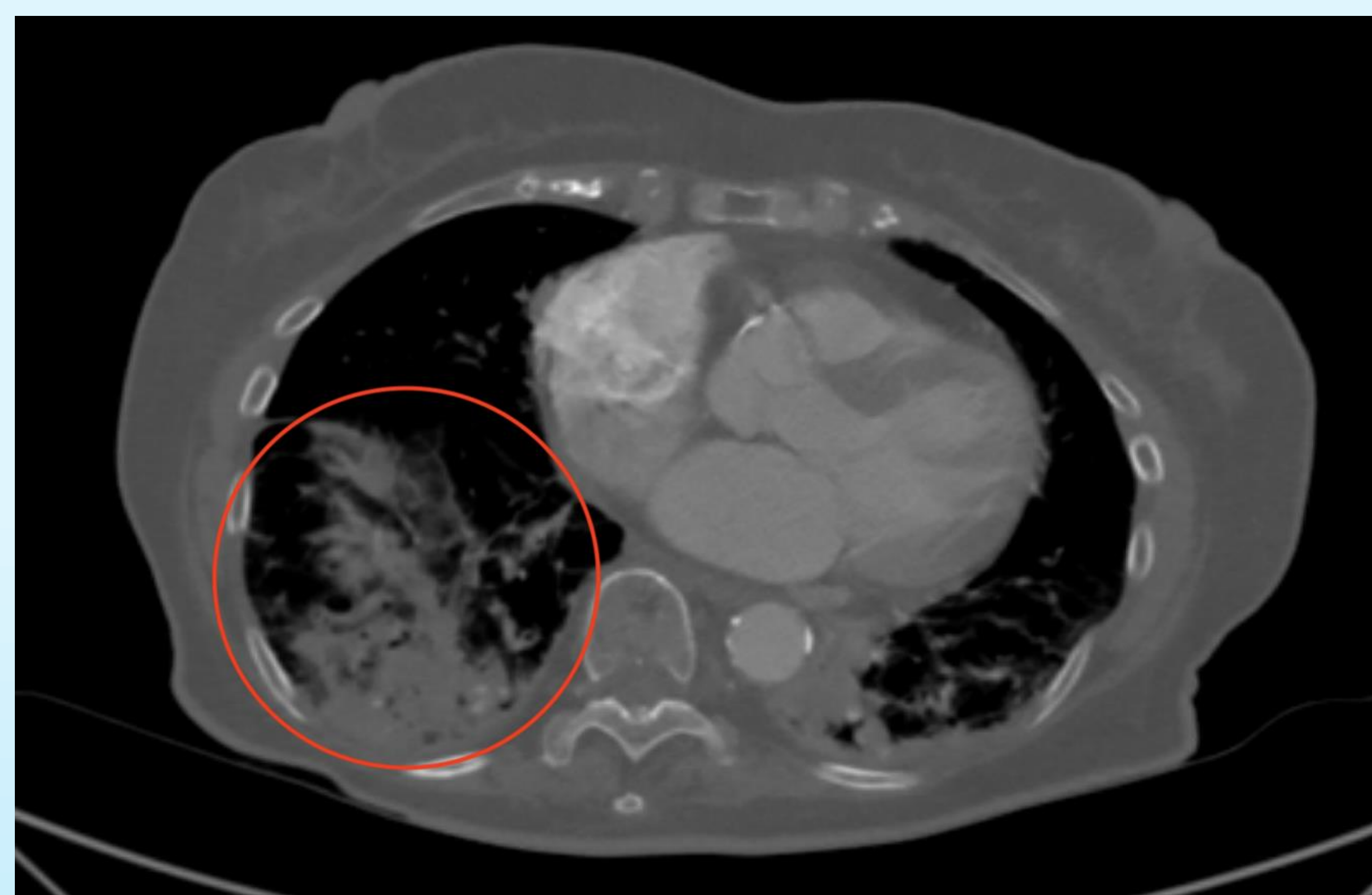


Figure 1: CT chest without contrast demonstrates moderate amount of dense consolidation and air bronchograms in right lower lobe (both circled in red), as well as in the left lower lobe

Case Presentation

A female in her 90s with a history of hypertension, hyperlipidemia, bladder prolapse, and glaucoma presented with two weeks of productive cough, generalized weakness, poor oral intake, and urinary hesitancy. She remained alert without neurological deficits. Initial vitals were notable for elevated blood pressure, and physical exam revealed signs of dehydration and respiratory distress. Labs showed leukocytosis, hyponatremia (Na 122), elevated lactic acid, and BNP. Imaging demonstrated bilateral lower lobe pneumonia with air bronchograms. Further workup showed elevated urine osmolality and sodium, low serum osmolality and uric acid, consistent with SIADH. Initial management with fluid restriction and salt tablets showed minimal response. On hospital day three, a single dose of tolvaptan increased serum sodium from 124 to 132 mEq/L. The patient improved clinically and was discharged with a short antibiotic course and plans for outpatient imaging to rule out an underlying lung mass.

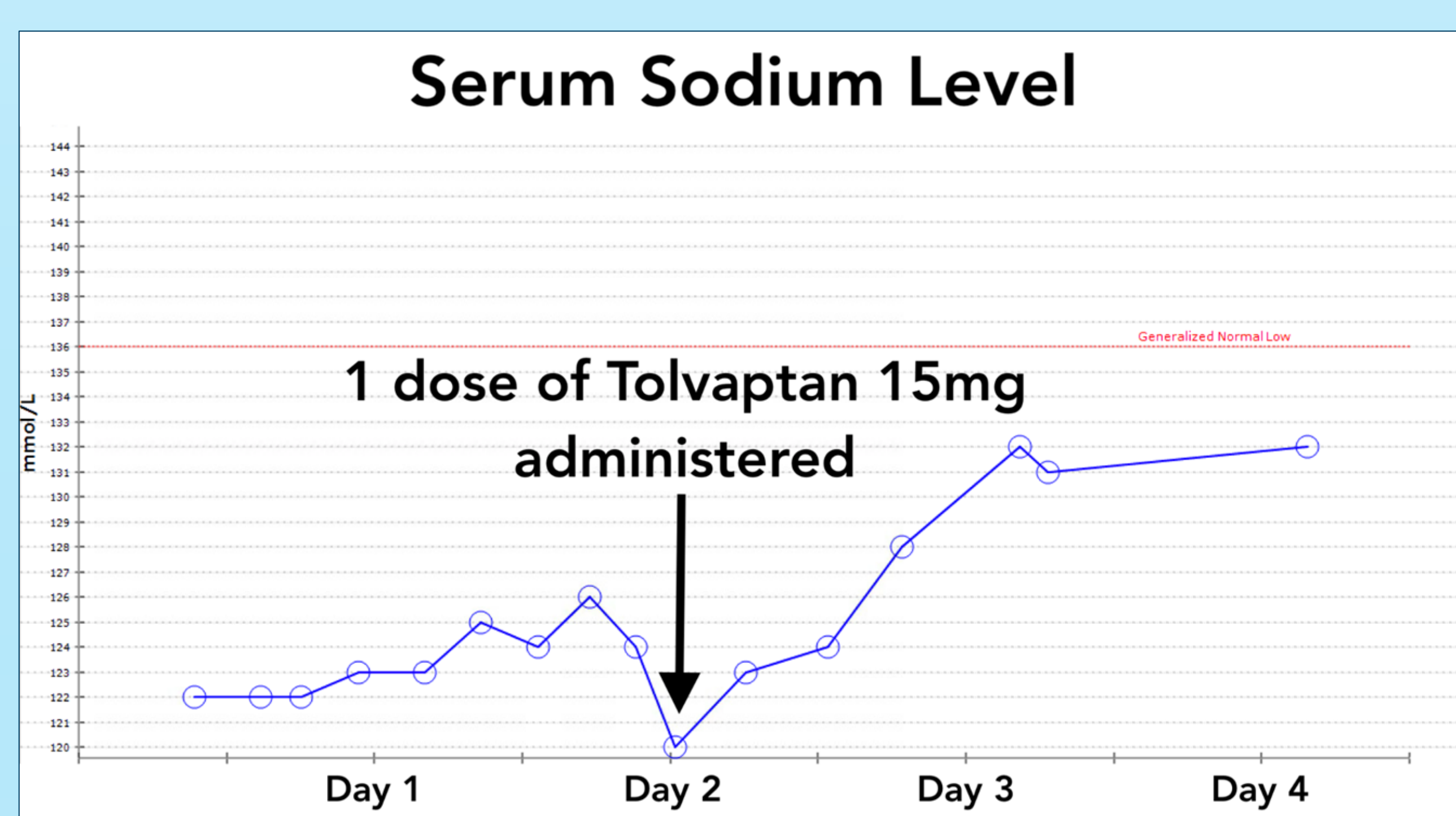


Figure 2: Serum Na level increases sharply after administration of a single 15mg dose of Tolvaptan

Discussion

The pathophysiology as to how pneumonia can cause SIADH is still being studied. However, several mechanisms have been proposed:

- Pneumonia causes lung injury, which leads to pulmonary vasoconstriction thus reducing left atrial filling causing decreased atrial receptor firing signaling the hypothalamus to increase ADH.
- Inflammatory cytokines such as Interleukin-6 (IL-6) and Interleukin-1b (IL-1b) have been shown to stimulate non-osmotic secretion of ADH as well damage the alveolar basement membrane, leading to pulmonary vasoconstriction, thus increasing ADH via the mechanism discussed above. While this mechanism has been extensively studied in COVID-19 pneumonia, studies are showing that it may play a significant role in CAP as well.
- Pressure on the diaphragm during pneumonia can trigger the release of ADH.

The use of tolvaptan requires careful monitoring to avoid overly rapid correction of sodium, which carries its own risks, particularly osmotic demyelination syndrome (ODS), therefore sodium should not be corrected more than 8-12 mEq/L in the first 24 hours. The primary treatment should focus on managing the pneumonia, which is essential in resolving SIADH as studies have shown that appropriate antibiotic treatment of the pneumonia alone can resolve the SIADH. Patients can be discharged once serum sodium levels have stabilized and symptoms have resolved; one dose of tolvaptan is sufficient in some cases, but continued monitoring and follow-up are crucial to ensure long-term stability and prevent recurrence of hyponatremia.

